

ARTI403: Image Processing

Lab 4: Intensity Transformations and Filtering: Spatial Domain

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Objective

To write a program that implements fundamental image processing algorithms.

Tools

Python 3, Jupyter, OpenCV, NumPy, Matplotlib, scikit-image.

Procedural Steps

Task 1: Thresholding

I used the camera image from scikit-image (the Parrot image from the manual is not available to me). I applied a fixed threshold with the values 0, 50, 100, 150 and 200. Pixels above the threshold become white (255) and the others become black (0).

```
for t in [0, 50, 100, 150, 200]:  
    ret, thresh = cv2.threshold(img, t, 255, cv2.THRESH_BINARY)
```

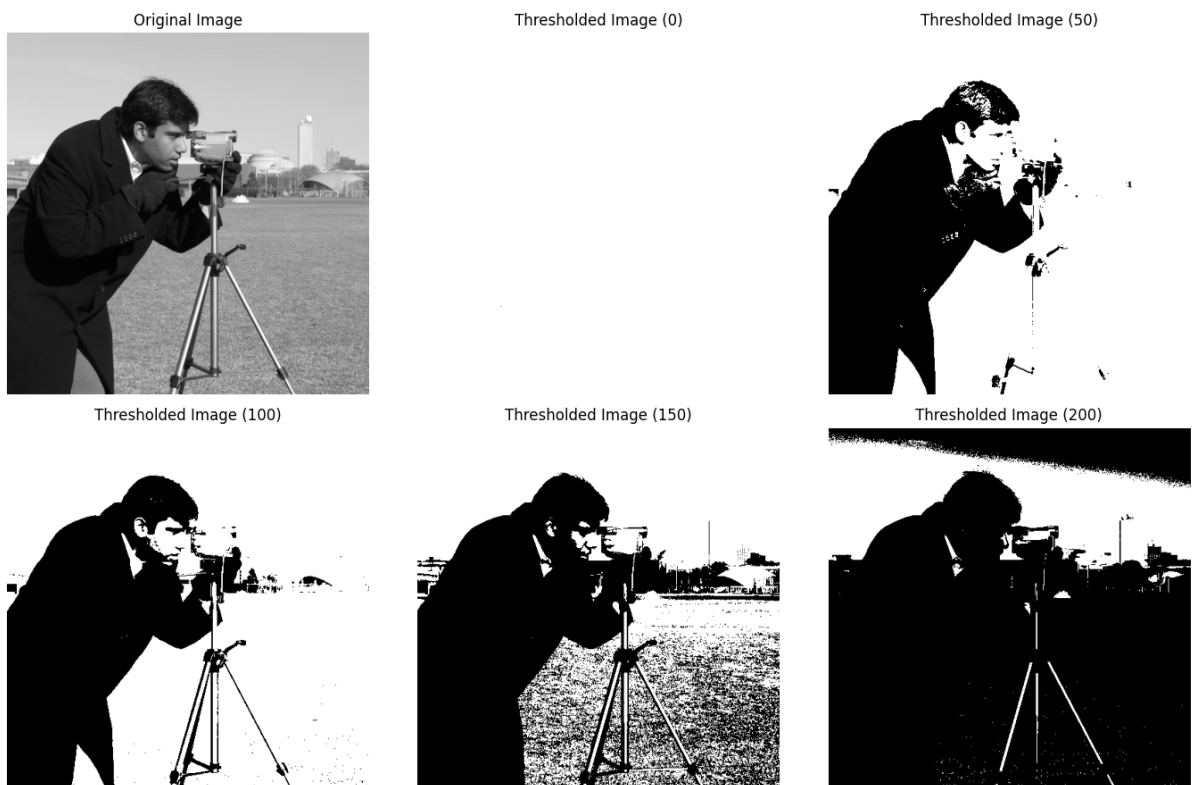


Figure 1: Original image and the thresholded images

When the threshold is higher, more pixels become black and only the brightest parts stay white.

Task 2: Histogram Processing

Steps 1 and 2: I imported the libraries and loaded the moon image (512 × 512, uint8).

```
img = data.moon()
```

Step 3: I did contrast stretching using the 2nd and 98th percentiles. The values were $p_2 = 78$ and $p_{98} = 129$.

```
p2, p98 = np.percentile(img, (2, 98))  
img_rescale = exposure.rescale_intensity(img, in_range=(p2, p98))
```

Step 4: I displayed the original image and the stretched image with their histograms.

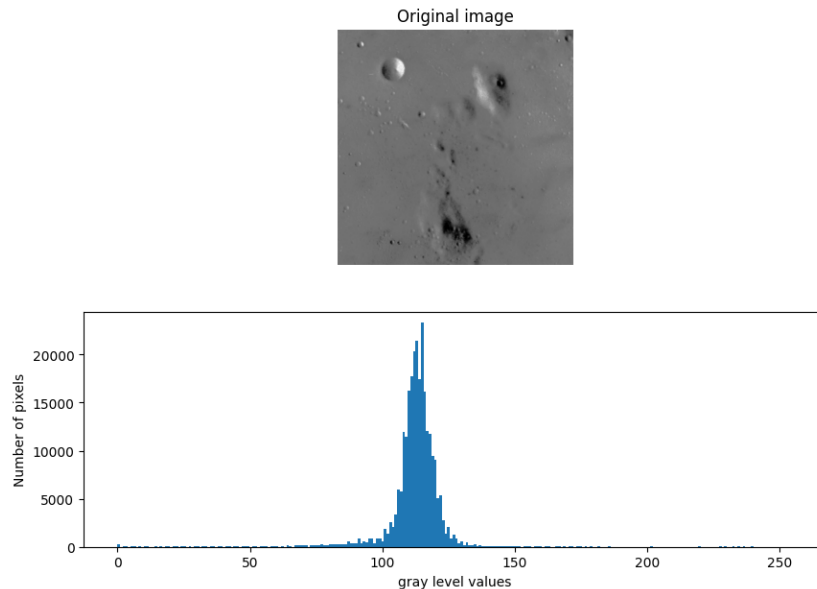


Figure 2: Original moon image and its histogram

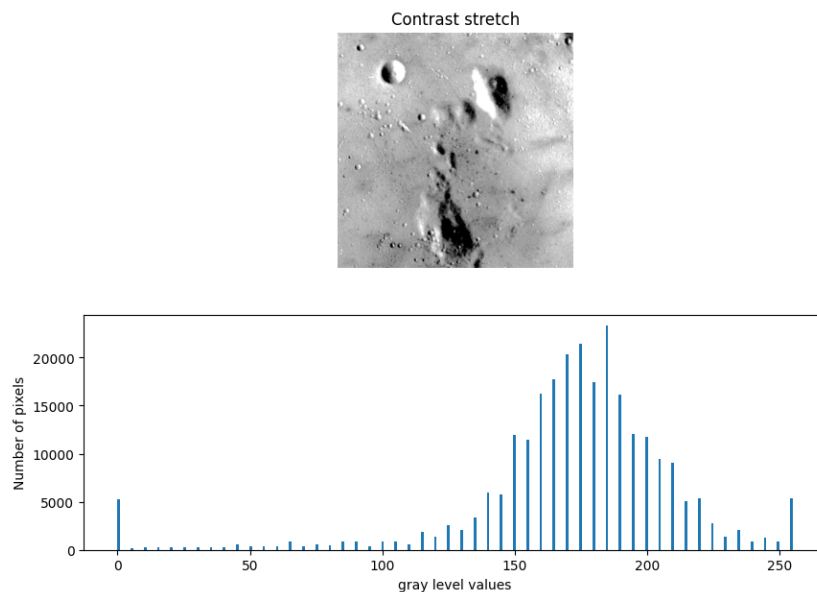


Figure 3: After contrast stretching (2nd to 98th percentile)

The histogram of the original image is narrow, so the image is low in contrast. After stretching, the gray values spread over the full range and the details are clearer.

Assessment

Task 1: Rescale intensity (3rd and 80th percentiles)

For the moon image I got $p_3 = 87$ and $p_{80} = 118$, then I rescaled the intensity and plotted the histogram.

```
p3, p80 = np.percentile(img, (3, 80))  
img_rescale_3_80 = exposure.rescale_intensity(img, in_range=(p3, p80))
```

Contrast stretch (3rd - 80th percentile)

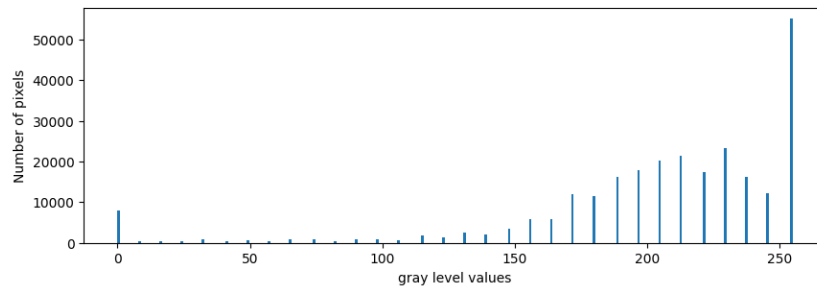


Figure 4: Contrast stretch (3rd to 80th percentile) and its histogram

Because I used a narrower range, more pixels reach black (0) or white (255), so the contrast is stronger than in the 2nd to 98th percentile stretch.

Task 2: Histogram equalization

I used `exposure.equalize_hist` on the moon image and displayed the image with its histogram. I also plotted the cumulative distribution before and after.

```
img_eq = exposure.equalize_hist(img)
```

Histogram equalized image

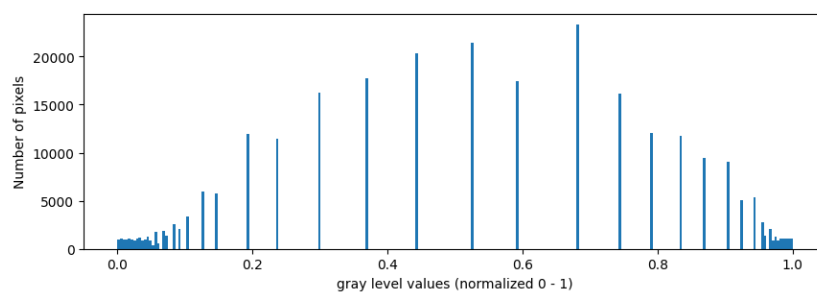


Figure 5: Equalized image and its histogram

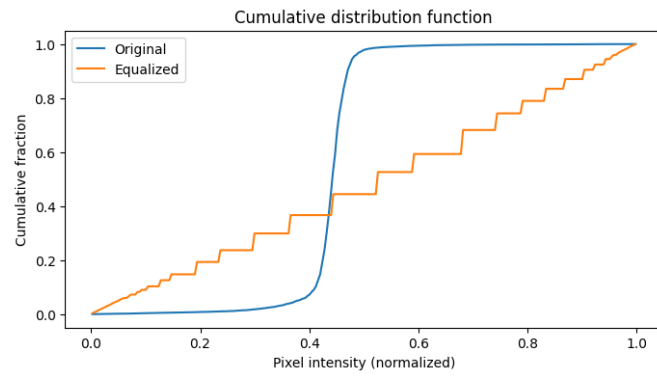


Figure 6: Cumulative distribution before and after equalization

After equalization the histogram is flatter and the cumulative distribution is close to a straight line.

Task 3: Histogram matching

I used the rocket image as the reference and the Chelsea image as the image to change, both from skimage.data. The matched image takes the colors of the rocket image.

```
reference = data.rocket()
image = data.chelsea()
matched = match_histograms(image, reference, channel_axis=-1)
```



Figure 7: Source, reference and matched images

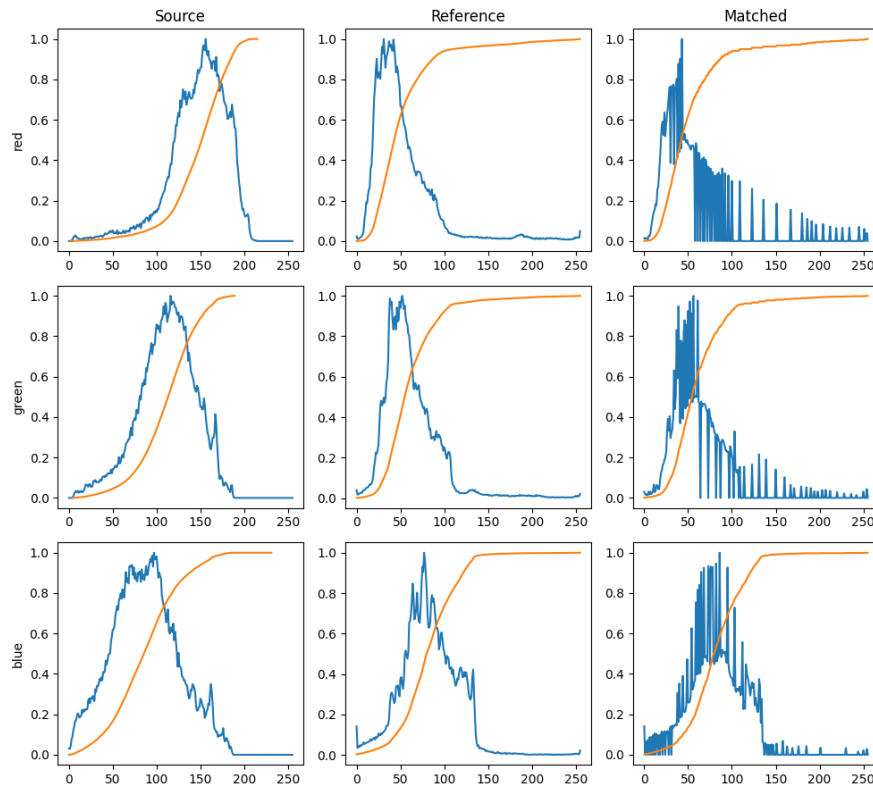


Figure 8: Histograms and CDFs of the R, G, B channels (source, reference, matched)

The histograms of the matched image look like the histograms of the reference image.